Coding Sample (R)

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This document presents the results of the Coding Sample (R). Task 1 examines the dynamic effects of democratic transitions on log GDP per capita using a country-year panel dataset. Task 2 develops and implements a data-audit strategy for a historical municipal elections dataset, focusing on internal consistency, OCR-related errors, and structured validation procedures.

Task 1 — Event Study / Dynamic Difference-in-Differences

1. Introduction

This task examines the effect of regime transitions on economic performance using a country-year panel with 184 countries observed between 1970 and 2010. The treatment variable is a binary indicator equal to one when a country is classified as a democracy and zero otherwise, and the outcome variable is log GDP per capita. The main empirical challenge is that treatment is non-absorbing: countries may transition into democracy and later revert to non-democracy, and some countries experience multiple switches over the sample period. This feature is central both substantively and econometrically.

The analysis proceeds in four steps. First, I motivate the estimator choice. Second, I present descriptive evidence on the panel structure and treatment variation. Third, I estimate a dynamic event study using the estimator of de Chaisemartin and D'Haultfœuille (2020, 2024), which is designed for settings with treatment switching and heterogeneous dynamic effects. Fourth, I construct a stacked difference-in-differences event study as a comparison exercise. Throughout, transitions into democracy and transitions out of democracy are estimated separately, since the underlying effects are expected to differ not only in sign but also in magnitude and timing.

2. Data and Descriptive Evidence

The dataset consists of a country-year panel containing a World Bank country code, calendar year, a democracy indicator, and log GDP per capita. From this panel, I construct lagged democracy status and two transition indicators: `trans_in`, equal to one when a country moves from non-democracy to democracy, and `trans_out`, equal to one when a country moves from democracy to non-democracy.

Before turning to the main estimates, it is useful to describe the aggregate evolution of the data. Figure 1 shows the share of countries classified as democracies over time. The figure displays a substantial upward trend, especially from the late 1980s onward, which is consistent with the well-known wave of democratization during the post-Cold War period. This pattern also makes clear that treatment timing is staggered rather than concentrated in a single period.

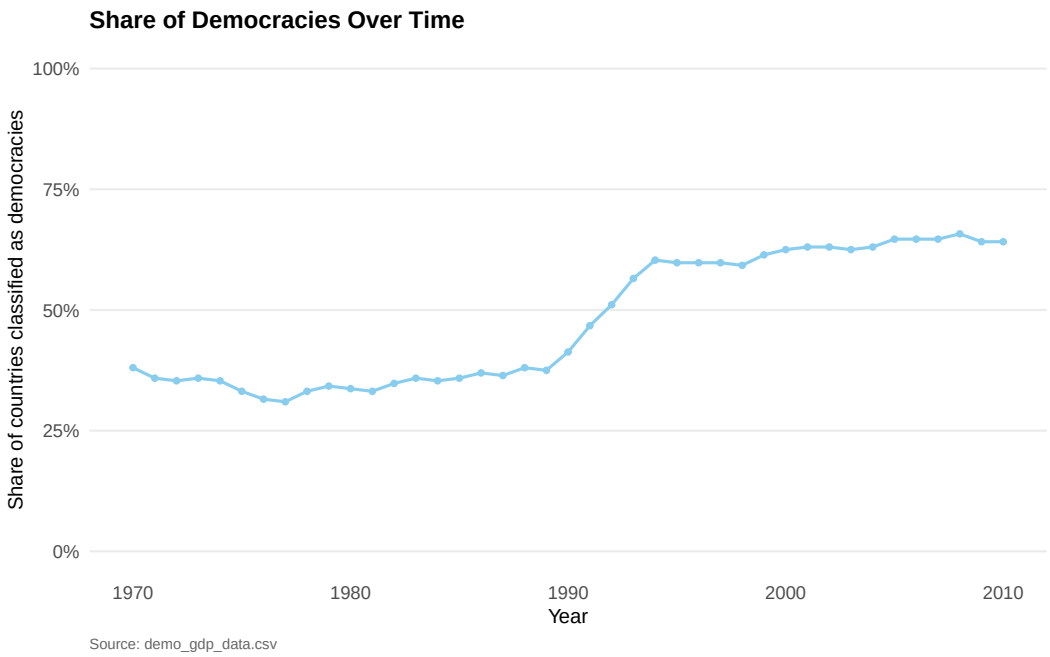

Figure 1: Share of Democracies Over Time

Figure 2 reports mean log GDP per capita by democracy status. Democracies are richer on average throughout the period. This comparison is purely descriptive and should not be interpreted causally, since countries that are democratic may differ systematically from non-democracies in many dimensions. Still, the figure is informative as a first pass and suggests that institutional regime type is associated with substantial cross-country differences in income levels.

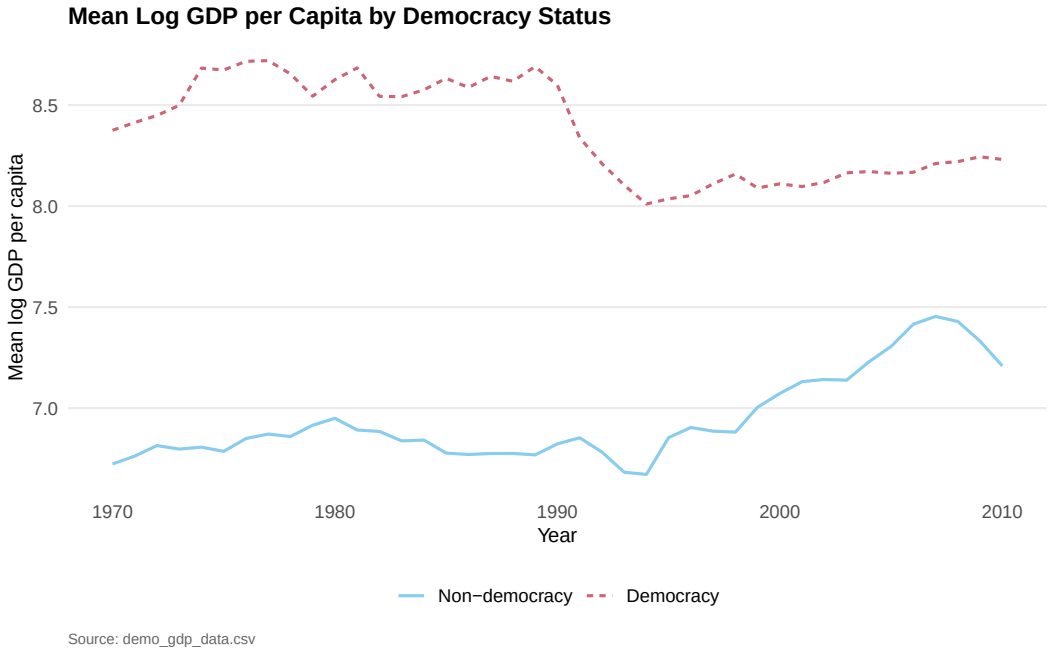

Figure 2: Mean Log GDP per Capita by Democracy Status

Figure 3 plots the distribution of first democratization years. The timing of first transitions is dispersed across the full sample horizon, again underscoring that this is a staggered-adoption environment. This feature is one of the reasons why simple two-period DID methods would be ill-suited to the application.

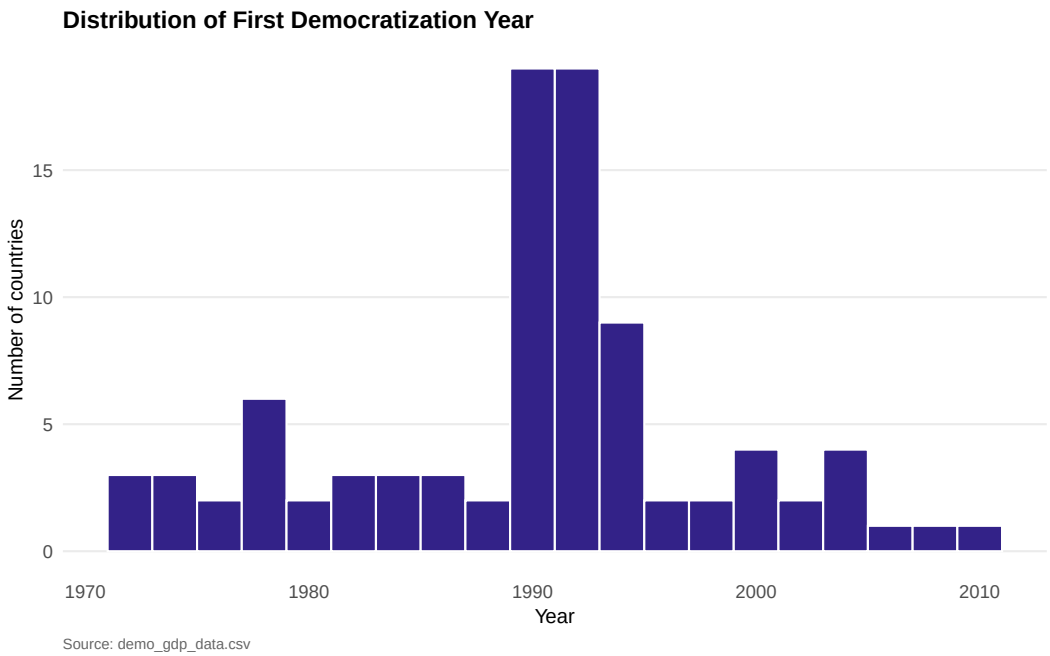

Figure 3: Distribution of First Democratization Year

Table 1 presents summary statistics for the main variables.

Table 1: Summary Statistics

Variable	N	Mean	SD	Min	Max
Log GDP per capita	6171	7.666	1.63	4.11	13.78
Democracy	7544	0.478	0.50	0.00	1.00

Country-year panel of 184 countries observed from 1970 to 2010 (41 years). Democracy is a binary indicator equal to 1 if the country is classified as a democracy. Log GDP per capita has 1,373 missing observations (78 countries with at least one missing year). 91 countries experience at least one transition into democracy; 19 of these also experience reversals.

Taken together, these descriptive statistics and figures suggest three key features of the data. First, democracy status evolves substantially over time. Second, treatment timing is highly staggered. Third, the sample includes both democratizations and democratic reversals. Any estimator used in this setting must therefore accommodate heterogeneity across cohorts and non-absorbing treatment dynamics.

3. Estimator Choice

A standard two-way fixed effects event-study estimator is not appropriate in this context. With staggered treatment timing and heterogeneous treatment effects, TWFE event-study regressions may assign negative weights and use already-treated units as controls for newly-treated units. These forbidden comparisons distort the causal interpretation of the event-time coefficients. The problem is even more severe here because treatment is non-absorbing: countries can move into and out of democracy multiple times, so treatment status is not monotone.

For this reason, the main specification uses the dynamic difference-in-differences estimator of de Chaisemartin and D'Haultfœuille (2020, 2024). This estimator is designed precisely for environments with treatment switching. It compares switchers to valid comparison units while avoiding the problematic weighting structure of TWFE under heterogeneous effects. In the present setting, that makes it the most appropriate baseline estimator.

I estimate transitions into democracy and transitions out of democracy separately. This separation is important for two reasons. First, the economic consequences of democratization and democratic breakdown are unlikely to be symmetric. Second, pooling the two directions would blur interpretation by averaging together effects that likely differ in sign, timing, and persistence.

As a complementary exercise, I also estimate a stacked event-study design following the logic of Cengiz et al. (2019). In this approach, each transition event defines a cohort-specific sub-experiment centered around the transition year, and the final stacked panel pools these sub-experiments. The stacked DID is not the main estimator, but it provides a transparent comparison design and a useful robustness check.

4. Main Results: de Chaisemartin and D'Haultfœuille Event Studies

I begin with the dynamic DID estimates based on de Chaisemartin and D'Haultfœuille. Standard errors are clustered at the country level. The estimator reports placebo coefficients for pre-transition periods and dynamic treatment effects for post-transition periods.

4.1 Transitions into Democracy

Figure 4 presents the event study for transitions into democracy. The placebo coefficients (periods -5 to -1) are positive and statistically significant, with a joint test of nullity rejecting at $p = 0.0006$. This indicates a clear violation of the parallel trends assumption: countries that democratize were on declining GDP trajectories relative to non-switchers in the years before the transition. The post-transition effects (periods 1 to 10) are negative, ranging from -0.019 to -0.056, but none is individually significant and the joint test fails to reject the null of zero effects ($p = 0.39$).

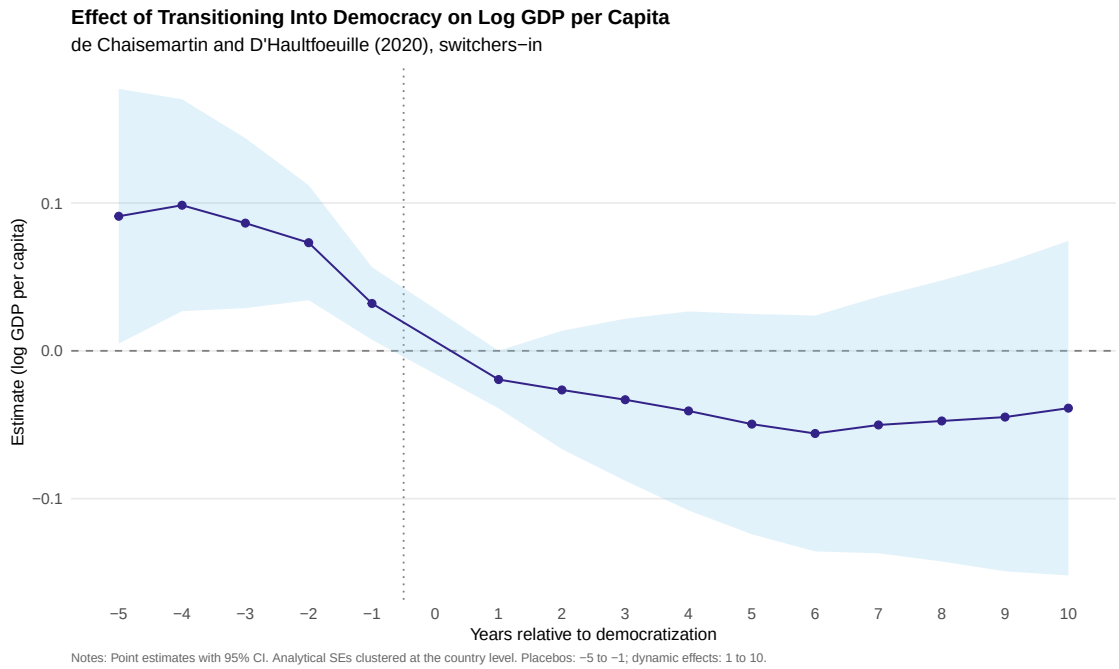

Figure 4: Effect of Transitioning Into Democracy on Log GDP per Capita: de Chaisemartin and D'Haultfœuille

The combination of significant pre-trends and insignificant post-treatment effects complicates causal interpretation. The negative post-treatment coefficients may simply reflect a continuation of the pre-existing relative decline rather than a causal effect of democratization. This pattern is consistent with the view that economic deterioration often precipitates regime change: countries tend to democratize during or after periods of poor economic performance relative to stable non-democracies.

4.2 Transitions out of Democracy

Figure 5 presents the corresponding event study for transitions out of democracy. Here the pattern is reversed. The placebo coefficients are negative (ranging from -0.025 to -0.139), with placebo periods -1 through -3 individually significant, though the joint test does not reject at conventional levels ($p = 0.16$), likely due to the small number of switchers (only 15–16 countries). The post-transition effects are positive and increasing over time, reaching $+0.152$ by period 10, with the joint test of effects marginally significant ($p = 0.052$).

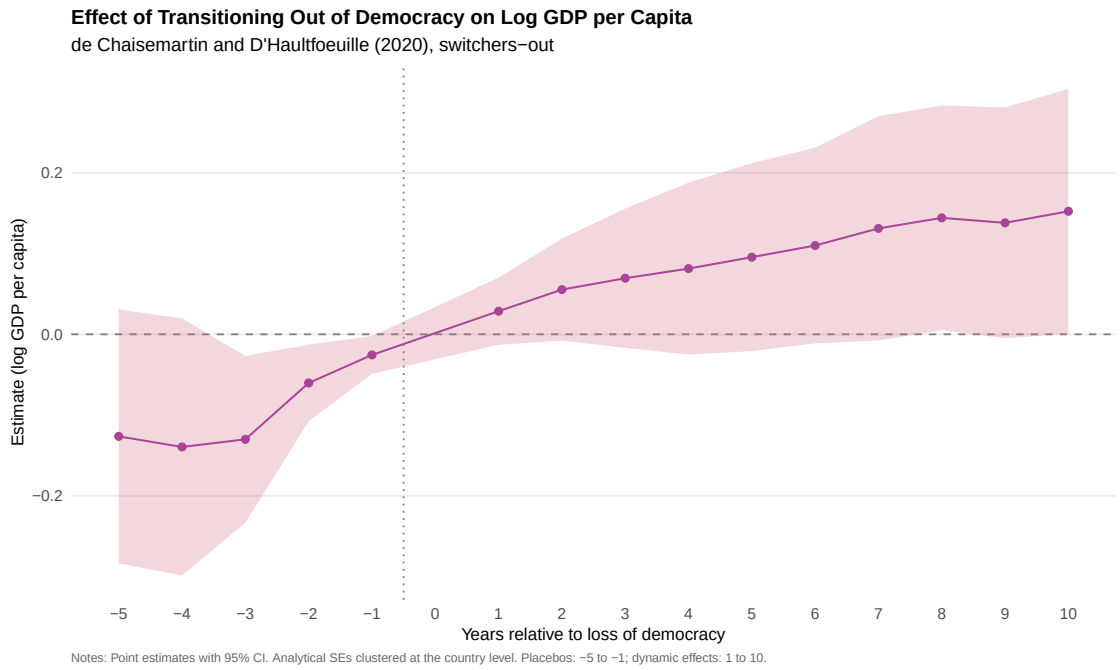

Figure 5: Effect of Transitioning Out of Democracy on Log GDP per Capita: de Chaisemartin and D'Haultfœuille

These positive post-transition coefficients indicate that countries losing democracy experience GDP *increases* relative to non-switchers. As with the switchers-in results, the pre-trend pattern suggests caution. The negative placebos imply that countries losing democracy were on improving GDP trajectories relative to controls before the reversal. This is consistent with the political economy intuition that democratic breakdowns may be more likely when incumbent elites have access to growing resources, or when economic performance reduces demand for democratic accountability. The small number of switchers (15–16 countries) also limits statistical power and the precision of these estimates.

4.3 Combined Plot

Figure 6 overlays the two sets of dynamic estimates. The pattern is visually striking in its symmetry: transitions into democracy are associated with negative post-treatment coefficients, while transitions out are associated with positive ones. Importantly, both directions exhibit pre-trend violations of opposite sign, suggesting that regime transitions in both directions occur at inflection points in countries' relative economic trajectories.

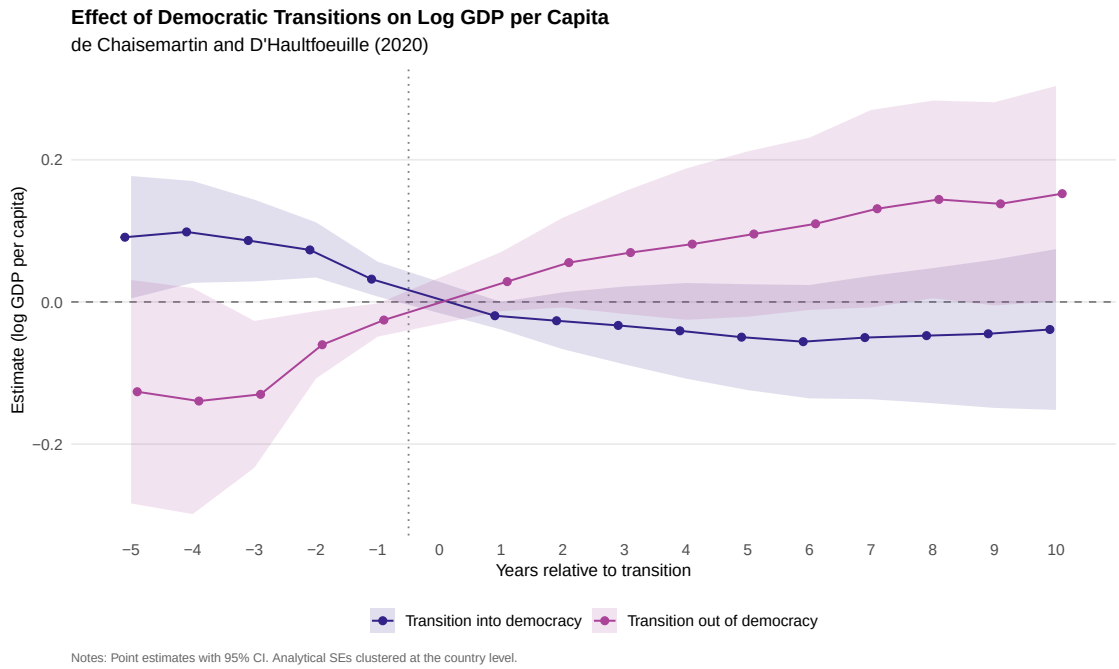

Figure 6: Effect of Democratic Transitions on Log GDP per Capita: Combined de Chaisemartin and D'Haultfœuille Estimates

For completeness, Tables 2 and 3 report the underlying numerical estimates.

Table 2: dCDH Estimates: Transitions Into Democracy

Type	Period	Estimate	Lower 95% CI	Upper 95% CI
Placebo	-5	0.0910	0.0049	0.1772
Placebo	-4	0.0985	0.0268	0.1701
Placebo	-3	0.0864	0.0289	0.1439
Placebo	-2	0.0731	0.0343	0.1120
Placebo	-1	0.0320	0.0075	0.0566
Effect	1	-0.0194	-0.0389	0.0001
Effect	2	-0.0265	-0.0665	0.0135
Effect	3	-0.0331	-0.0879	0.0217
Effect	4	-0.0407	-0.1079	0.0266
Effect	5	-0.0496	-0.1241	0.0249
Effect	6	-0.0559	-0.1357	0.0238
Effect	7	-0.0501	-0.1369	0.0366
Effect	8	-0.0474	-0.1425	0.0476
Effect	9	-0.0448	-0.1491	0.0595
Effect	10	-0.0388	-0.1519	0.0744

Estimates from the de Chaisemartin and D'Haultfœuille (2020, 2024) estimator with switchers = "in". Analytical standard errors clustered at the country level. 67 switchers. Joint test of effects: $p = 0.3878$. Joint test of placebos: $p = 6e - 04$.

Table 3: dCDH Estimates: Transitions Out of Democracy

Type	Period	Estimate	Lower 95% CI	Upper 95% CI
Placebo	-5	-0.1263	-0.2835	0.0309
Placebo	-4	-0.1394	-0.2985	0.0196
Placebo	-3	-0.1300	-0.2332	-0.0268
Placebo	-2	-0.0602	-0.1077	-0.0128
Placebo	-1	-0.0255	-0.0489	-0.0021
Effect	1	0.0286	-0.0130	0.0702
Effect	2	0.0553	-0.0078	0.1185
Effect	3	0.0694	-0.0168	0.1557
Effect	4	0.0813	-0.0251	0.1878
Effect	5	0.0955	-0.0208	0.2118
Effect	6	0.1099	-0.0113	0.2310
Effect	7	0.1311	-0.0078	0.2700
Effect	8	0.1442	0.0051	0.2833
Effect	9	0.1380	-0.0050	0.2809
Effect	10	0.1523	0.0007	0.3038

Estimates from the de Chaisemartin and D'Haultfœuille (2020, 2024) estimator with switchers = "out". Analytical standard errors clustered at the country level. 16 switchers. Joint test of effects: $p = 0.0524$. Joint test of placebos: $p = 0.1612$.

5. Stacked Difference-in-Differences Event Study

I next construct a stacked DID event study as a comparison exercise. For each transition event occurring in year g, I define a cohort-specific window of $[g-10, g+10]$. The treated unit is the country that experiences the transition. The control group consists of countries with no regime switch of any kind inside that event window. Each sub-experiment is then stacked together, and the model is estimated with stack-by-unit and stack-by-year fixed effects.

This design is attractive because it creates transparent comparisons using clean controls. At the same time, identification rests on a cohort-specific parallel trends assumption: absent the transition, treated countries and clean controls would have followed parallel trajectories within the event window. Unlike the de Chaisemartin and D'Haultfœuille estimator, the stacked design depends directly on the chosen window and on the sample restrictions used to define uncontaminated controls.

5.1 Transitions into Democracy

Figure 7 shows the stacked DID estimates for transitions into democracy. The pre-treatment coefficients are positive and significant across the full window (-10 to -2), confirming the pre-trend pattern observed in the main estimator. Post-treatment coefficients are negative and statistically significant in the early post-transition years (periods 0 through 7), with point estimates ranging from -0.028 to -0.061. The stacked DID thus corroborates the dCDH finding that democratization is not followed by relative GDP gains in this specification, and that the result is confounded by pre-existing differential trends.

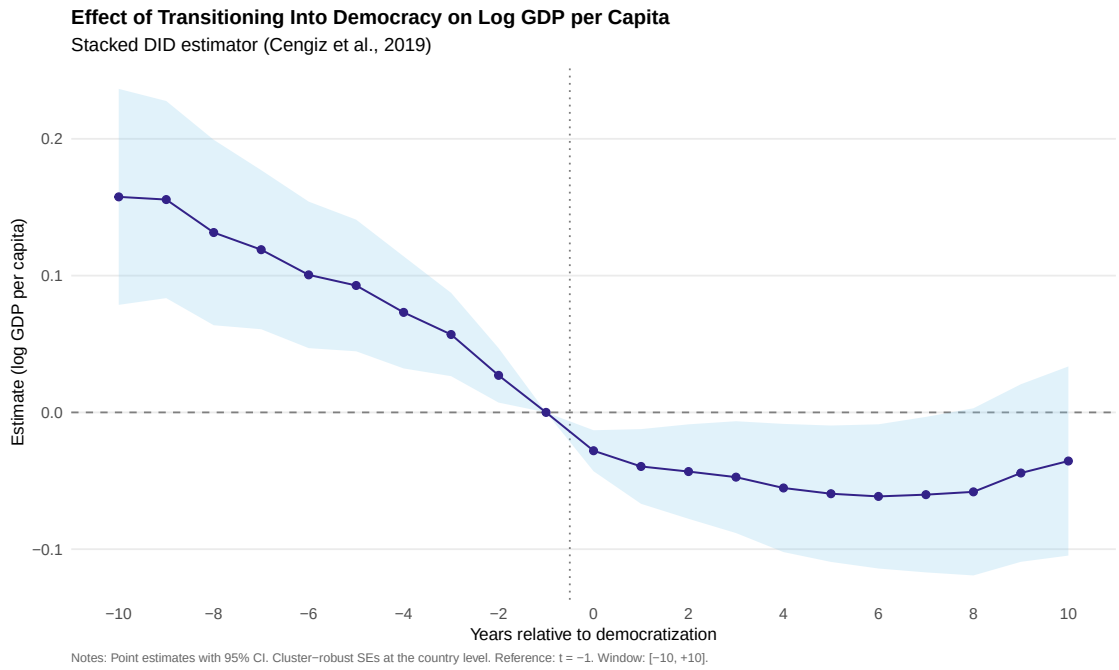

Figure 7: Stacked DID Event Study: Transitions Into Democracy

5.2 Transitions out of Democracy

Figure 8 shows the stacked DID estimates for transitions out of democracy. As with transitions in, the pre-treatment coefficients are positive and significant, indicating that countries losing democracy were on higher relative GDP paths before the reversal. Post-treatment coefficients are negative and significant (periods 0 through 7), with estimates between -0.036 and -0.088. Notably, the stacked DID produces negative post-treatment effects for democratic reversals, whereas the dCDH estimator produces positive ones. This sign discrepancy is discussed in the comparison section below.

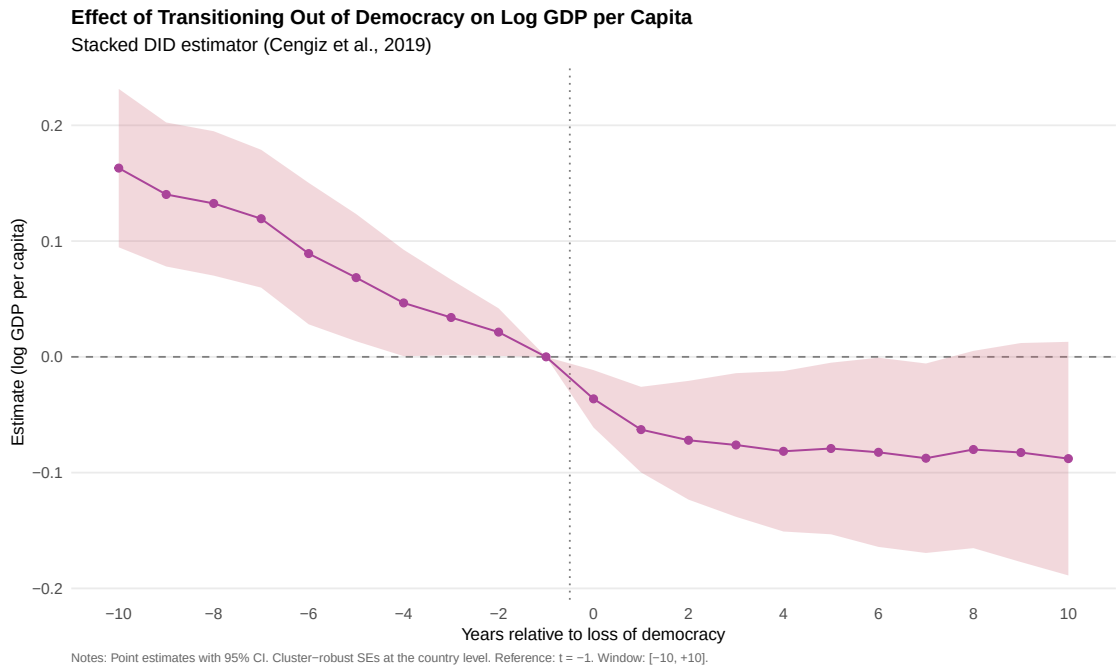

Figure 8: Stacked DID Event Study: Transitions Out of Democracy

Tables 4 and 5 present the corresponding estimates.

Table 4: Stacked DID Estimates: Transitions Into Democracy

Type	Period	Estimate	SE	Lower 95% CI	Upper 95% CI
Pre-treatment	-10	0.1576	0.0403	0.0786	0.2365
Pre-treatment	-9	0.1556	0.0368	0.0836	0.2277
Pre-treatment	-8	0.1315	0.0346	0.0637	0.1993
Pre-treatment	-7	0.1190	0.0297	0.0608	0.1771
Pre-treatment	-6	0.1006	0.0273	0.0470	0.1541
Pre-treatment	-5	0.0928	0.0246	0.0447	0.1409
Pre-treatment	-4	0.0731	0.0209	0.0321	0.1141
Pre-treatment	-3	0.0569	0.0155	0.0265	0.0874
Pre-treatment	-2	0.0271	0.0102	0.0072	0.0470
Pre-treatment	-1	0.0000	0.0000	0.0000	0.0000
Treatment	0	-0.0280	0.0076	-0.0430	-0.0130
Post-treatment	1	-0.0395	0.0139	-0.0668	-0.0122
Post-treatment	2	-0.0433	0.0176	-0.0778	-0.0087
Post-treatment	3	-0.0473	0.0209	-0.0882	-0.0065
Post-treatment	4	-0.0553	0.0239	-0.1021	-0.0084
Post-treatment	5	-0.0595	0.0254	-0.1093	-0.0096
Post-treatment	6	-0.0614	0.0269	-0.1141	-0.0087
Post-treatment	7	-0.0601	0.0290	-0.1170	-0.0033
Post-treatment	8	-0.0581	0.0312	-0.1192	0.0030
Post-treatment	9	-0.0443	0.0331	-0.1092	0.0206
Post-treatment	10	-0.0355	0.0353	-0.1047	0.0336

Stacked DID following Cengiz et al. (2019). Each transition event defines a cohort-specific sub-experiment with event window $[-10, +10]$. Controls have no regime switch of any kind inside the window. Cluster-robust standard errors at the country level. Reference period: $t = -1$.

Table 5: Stacked DID Estimates: Transitions Out of Democracy

Type	Period	Estimate	SE	Lower 95% CI	Upper 95% CI
Pre-treatment	-10	0.1631	0.0349	0.0946	0.2315
Pre-treatment	-9	0.1403	0.0317	0.0781	0.2025
Pre-treatment	-8	0.1325	0.0318	0.0701	0.1949
Pre-treatment	-7	0.1193	0.0304	0.0598	0.1789
Pre-treatment	-6	0.0892	0.0312	0.0281	0.1503
Pre-treatment	-5	0.0684	0.0280	0.0135	0.1234
Pre-treatment	-4	0.0466	0.0235	0.0006	0.0925
Pre-treatment	-3	0.0340	0.0167	0.0013	0.0667
Pre-treatment	-2	0.0214	0.0105	0.0008	0.0419
Pre-treatment	-1	0.0000	0.0000	0.0000	0.0000
Treatment	0	-0.0363	0.0127	-0.0611	-0.0115
Post-treatment	1	-0.0628	0.0189	-0.0998	-0.0259
Post-treatment	2	-0.0720	0.0262	-0.1233	-0.0207
Post-treatment	3	-0.0761	0.0317	-0.1382	-0.0141
Post-treatment	4	-0.0816	0.0354	-0.1509	-0.0123
Post-treatment	5	-0.0792	0.0378	-0.1533	-0.0050
Post-treatment	6	-0.0824	0.0417	-0.1642	-0.0007
Post-treatment	7	-0.0876	0.0418	-0.1694	-0.0057
Post-treatment	8	-0.0800	0.0435	-0.1652	0.0052
Post-treatment	9	-0.0826	0.0483	-0.1772	0.0119
Post-treatment	10	-0.0879	0.0515	-0.1888	0.0130

Stacked DID following Cengiz et al. (2019). Each transition event defines a cohort-specific sub-experiment with event window $[-10, +10]$. Controls have no regime switch of any kind inside the window. Cluster-robust standard errors at the country level. Reference period: $t = -1$.

6. Comparing the Two Approaches

The final step is to compare the main dynamic DID estimator with the stacked DID design. Figures 9 and 10 place the two sets of estimates side by side for transitions into and out of democracy, respectively.

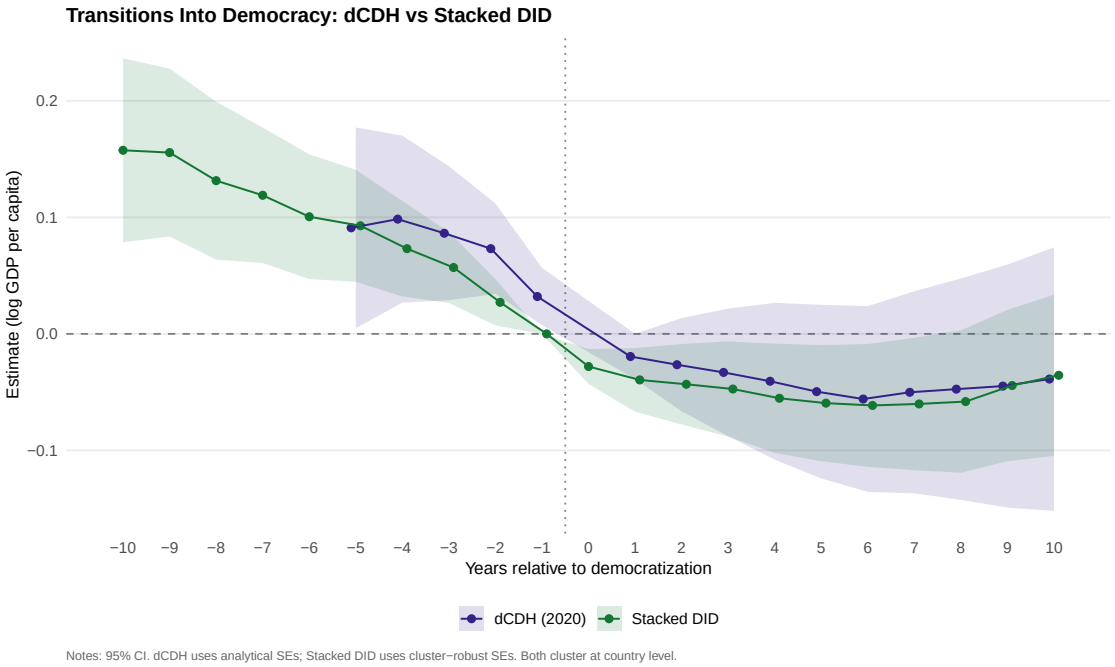

Figure 9: Transitions Into Democracy: de Chaisemartin and D'Haultfœuille versus Stacked DID

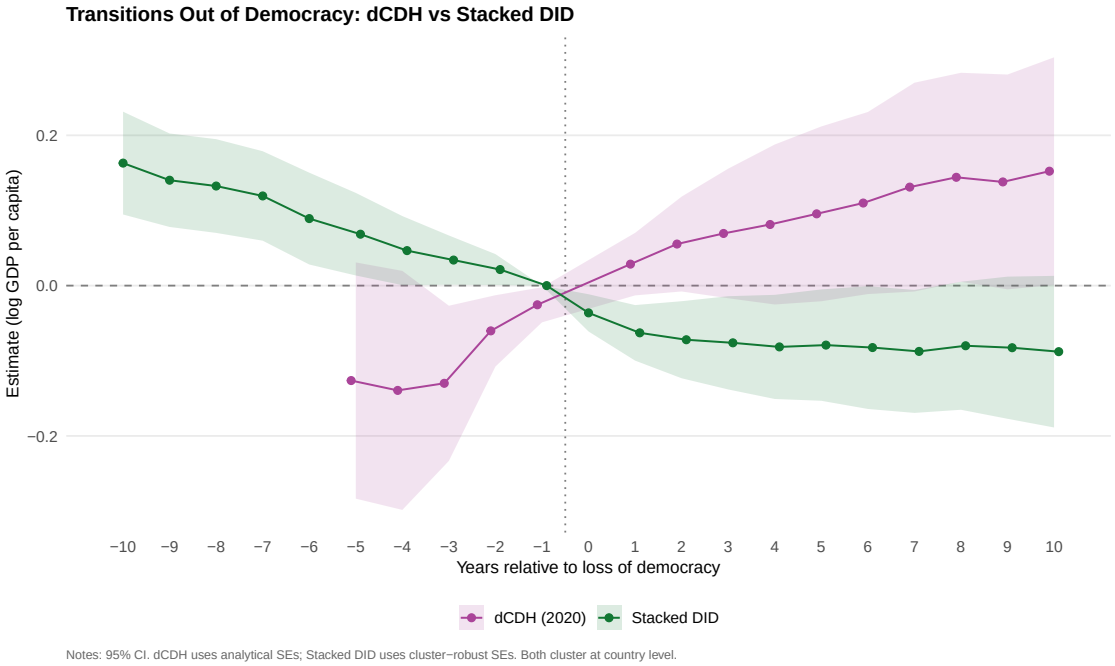

Figure 10: Transitions Out of Democracy: de Chaisemartin and D'Haultfœuille versus Stacked DID

The comparison between the two estimators reveals both agreement and disagreement. For transitions into democracy (Figure 9), the two approaches are qualitatively aligned: both show positive pre-trends and negative post-treatment coefficients, although the stacked DID estimates are more precise and statistically significant in the early post-treatment periods.

For transitions out of democracy (Figure 10), the two estimators disagree on the sign of the post-treatment effects. The dCDH estimator shows positive effects (GDP rises after losing democracy), while the stacked DID shows negative effects. This discrepancy likely reflects differences in how the two methods handle pre-existing trends and define comparison groups. The dCDH estimator compares switchers to non-switchers with the same baseline treatment, adjusting for contemporaneous trends. The stacked DID, by contrast, uses only clean controls within a fixed event window, and the strong positive pre-trends in the stacked sample may contaminate the post-treatment estimates through a continuation of the differential trend. With only 15–16 switchers-out in the dCDH and a similarly small treated sample in the stacked design, both sets of estimates should be interpreted with caution.

The disagreement underscores a broader point: in settings with violated parallel trends and non-absorbing treatment, the choice of estimator and comparison group materially affects the conclusions. Neither approach can deliver a clean causal estimate without a credible parallel trends assumption, and the pre-trend violations documented here suggest that this assumption is unlikely to hold unconditionally.

7. Discussion and Limitations

The most important finding of this analysis is that the parallel trends assumption is violated for transitions into democracy ($p = 0.0006$ on the joint placebo test). Countries that democratize were on declining GDP trajectories relative to non-switchers before the transition occurred. This pre-existing divergence makes it difficult to attribute the post-treatment GDP path causally to the regime change itself.

This pattern is substantively informative even if it prevents clean causal identification. It suggests that democratization is partly endogenous to economic conditions: countries tend to transition to democracy during periods of relative economic weakness. This is consistent with the political economy literature on the determinants of regime change, which emphasizes the role of economic crises in triggering democratic transitions (Acemoglu and Robinson, 2006; Haggard and Kaufman, 2016).

For transitions out of democracy, the small number of switchers (15–16) limits both power and the reliability of pre-trend tests. The dCDH and stacked DID estimators disagree on the sign of post-treatment effects, further cautioning against strong causal claims.

Relative to the findings of Acemoglu et al. (2019), who report positive long-run effects of democracy on GDP, the present analysis uses a shorter panel (1970–2010 versus 1960–2010), no covariates, and a different estimator. Acemoglu et al. control for lags of GDP and use an instrumental variable strategy based on regional democratization waves. These additional controls and identification strategies may account for the pre-existing differential trends that confound the unconditional event study presented here. The results in this report should therefore be read as illustrating the importance of addressing pre-trends and endogeneity, rather than as evidence against the democracy-growth relationship.

Two additional caveats deserve mention. First, the dCDH estimator requires a no-carryover assumption: potential outcomes depend only on current treatment status, not on the full history of regime changes. This assumption is likely restrictive in the present context, since past democratization may have lasting effects on institutions, human capital, and investment even after a reversal. Second, the stacked DID relies on the availability of clean controls — countries with no regime change of any kind within the event window — and the composition of this control group may differ systematically from the treated countries in ways that affect the estimates.

8. Conclusion

This report estimates the dynamic effect of democratic transitions on log GDP per capita using a country-year panel with non-absorbing treatment. The preferred estimator is the dynamic DID approach of de

Chaisemartin and D'Haultfœuille (2020, 2024), with a stacked DID event study as a comparison.

The central finding is that parallel trends is violated for transitions into democracy: countries that democratize were already on declining relative GDP trajectories before the regime change. Post-treatment effects are negative but statistically insignificant. For transitions out of democracy, the evidence is mixed, with the two estimators disagreeing on the sign of post-treatment effects, and the small number of switchers limiting statistical power.

These results highlight the endogeneity of regime transitions to economic conditions and underscore the difficulty of identifying the causal effect of democracy on growth without additional controls or instruments. The findings are consistent with the broader methodological message of the recent DID literature: in settings with staggered and non-absorbing treatment, estimator choice and pre-trend diagnostics are essential for credible inference.

References

- Acemoglu, D. and Robinson, J.A. (2006). *Economic Origins of Dictatorship and Democracy.* Cambridge University Press.
- Acemoglu, D., Naidu, S., Restrepo, P. and Robinson, J.A. (2019). "Democracy Does Cause Growth." *Journal of Political Economy*, 127(1), 47–100.
- Callaway, B. and Sant'Anna, P.H.C. (2021). "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, 225(2), 200–230.
- Cengiz, D., Dube, A., Lindner, A. and Zipperer, B. (2019). "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics*, 134(3), 1405–1454.
- de Chaisemartin, C. and D'Haultfœuille, X. (2020). "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review*, 110(9), 2964–2996.
- de Chaisemartin, C. and D'Haultfœuille, X. (2024). "Difference-in-Differences Estimators of Intertemporal Treatment Effects." *Review of Economics and Statistics*, 1–45.
- Gardner, J. (2022). "Two-Stage Differences in Differences." Working Paper.
- Goodman-Bacon, A. (2021). "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics*, 225(2), 254–277.
- Haggard, S. and Kaufman, R.R. (2016). *Dictators and Democrats: Masses, Elites, and Regime Change.* Princeton University Press.
- Roth, J., Sant'Anna, P.H.C., Bilinski, A. and Poe, J. (2023). "What's Trending in Difference-in-Differences?" *Journal of Econometrics*, 235, 2218–2244.
- Sun, L. and Abraham, S. (2021). "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics*, 225(2), 175–199.

Task 2 — Data Audit Plan for Historical Elections Data

1. Audit Strategy

The objective of this audit is to assess the internal consistency and reliability of a municipality-election dataset digitized via OCR from historical paper records. Given the nature of OCR-generated data, errors may range from minor formatting inconsistencies to substantive misreporting of numeric values. The audit strategy therefore prioritizes checks according to their potential impact on data validity.

The first priority is to identify logical impossibilities and accounting inconsistencies. These include cases where the number of voters exceeds the number of registered electors, mismatches between total valid votes and the sum of party-level votes, inconsistencies involving invalid and blank ballots, and discrepancies between total council seats and their allocation across parties. These checks are essential because they identify observations that cannot be internally reconciled and therefore compromise the integrity of the dataset.

The second priority is to identify duplication and geographic inconsistencies. Because the dataset spans multiple elections across municipalities, the relevant unit of observation is the municipality-election pair. The audit therefore checks for exact and near-duplicate records, as well as inconsistencies between municipality names and standardized municipality codes. These issues affect the uniqueness and comparability of observations.

The final priority is to identify OCR-related formatting issues and structural missingness. These include capitalization inconsistencies, irregular spacing, and missing values in ballot-detail variables. While these issues are typically less consequential for inference, they are important for data harmonization and reproducibility.

Automatic corrections are restricted to formatting issues that can be addressed mechanically with high confidence. All discrepancies involving numeric values, accounting identities, duplication, or geographic identifiers are flagged for manual review.

2. Implemented Audit Checks

The audit is implemented as a reproducible pipeline that reads the raw dataset, applies a sequence of validation checks, and records all flagged issues in a structured audit log. Each check is defined programmatically and contributes one or more entries to the audit log, which includes the row identifier, the audit category, the specific check performed, a severity classification, and a recommended action.

The implemented checks cover five main dimensions. First, vote-logic checks verify whether turnout exceeds the number of registered electors and whether turnout rates are implausibly extreme. Second, vote-accounting checks compare reported totals with the sum of component vote variables. Third, seat-accounting checks compare total seats with the sum of party seat allocations. Fourth, duplication and geographic checks identify exact and near-duplicate records and mismatches between municipality names and standardized codes. Fifth, OCR-formatting checks detect capitalization, spacing, and name standardization issues.

All checks are implemented using a consistent structure: each validation rule filters problematic observations and appends them to a unified audit log with standardized fields. This ensures that the audit process is fully reproducible and that all issues are tracked in a transparent manner. The same logic is applied across all audit dimensions, and the resulting audit log is exported as a structured file for validation and documentation.

3. Issue Types and Proposed Actions

Each identified issue is associated with a recommended treatment based on its severity and the degree of ambiguity involved. Logical impossibilities, such as cases where turnout exceeds the number of registered electors, are classified as critical and flagged for immediate manual review. These cases indicate either OCR errors in key numeric fields or inconsistencies in the original records and cannot be corrected automatically without external validation.

High-severity issues include vote-accounting mismatches, seat-accounting inconsistencies, and near-duplicate observations. These issues may reflect misalignment between totals and their components, duplication of records, or partial transcription errors. Given their potential impact on empirical analysis, they are systematically flagged for manual inspection and correction.

Medium-severity issues include inconsistencies between municipality names and standardized municipality codes, as well as missing totals that prevent internal validation. These issues are often ambiguous: they may arise from OCR errors, administrative changes in municipality identifiers, or incomplete records. As a result, they require cross-validation with external sources or historical documentation before any correction is applied.

Low-severity issues are primarily related to formatting inconsistencies and OCR noise. These include capitalization errors, irregular spacing, and minor variations in municipality names. Such issues can be corrected automatically with high confidence using standardized string-cleaning procedures, provided that the original values are preserved for traceability. Structural missingness in ballot-detail variables is treated as a documented feature of the dataset rather than an error, and coding conventions are recorded but not altered.

4. Structured Audit Output

The audit produces a structured output file that records all identified issues at the observation level. Each entry corresponds to a specific validation failure and includes a unique row identifier, the audit category, the name of the check, a severity classification, and a recommended action.

This structure ensures that the audit process is transparent, reproducible, and easily interpretable. By separating detection from correction, the approach allows for systematic validation of flagged observations before any modification is made to the dataset. Recommended actions are standardized into categories such as `manual_review`, `manual_review_name`, `auto_correct`, and `note`, which facilitates downstream processing and documentation.

5. Audit Results

The audit identifies a total of 781 flags across 577 observations, indicating that each observation is flagged at least once. This high coverage is largely driven by low-severity issues, particularly structural missingness in ballot-detail variables and coding conventions related to invalid and blank ballots.

Table 6: Audit Overview

Metric	Value
Total flags	781
Unique rows flagged	577
Total rows in dataset	577

Critical inconsistencies are rare, with only a small number of cases where the number of voters exceeds the number of registered electors. High-severity issues are concentrated in vote-accounting mismatches, seat-accounting inconsistencies, and near-duplicate observations. Medium-severity issues are primarily associated with municipality-name and municipality-code mismatches and missing totals that limit internal validation.

Low-severity issues account for the majority of flags and are largely cosmetic or structural in nature. These findings suggest that while the dataset contains systematic OCR-related noise, the most consequential inconsistencies are limited to a relatively small subset of observations.

Table 7: Data Audit Summary: Flagged Issues by Category and Severity

Category	Check	Severity	Action	N
Vote_logic	votanti_gt_elettori	critical	manual_review	2
Duplicates	near_duplicate	high	manual_review	10
Seat_accounting	seat_total_mismatch	high	manual_review	7
Vote_accounting	vote_total_mismatch	high	manual_review	30
Vote_accounting	listvote_total_mismatch	high	manual_review	9
Geography	province_code_mismatch	medium	manual_review	2
Missing	missing_totals	medium	manual_review	21
OCR_name	name_code_mismatch	medium	manual_review_name	77
Vote_logic	extreme_turnout	medium	manual_review	3
Missing	structural_na_ballot_detail	low	note_structural_missing	442
OCR_name	preposition_capitalization	low	auto_correct_prepositions	46
OCR_name	all_caps_comune	low	auto_correct_to_title_case	26
OCR_name	spurious_spaces	low	auto_correct_merge_words	1
Vote_accounting	schede_bianche_in_invalidi	low	note_coding_convention	105

Severity levels: critical (logically impossible values), high (accounting identity failures or duplicates), medium (plausible but suspicious values requiring review), low (cosmetic or structural issues correctable automatically). N = number of flagged observations per check.

6. Conclusion

The proposed audit strategy provides a structured and reproducible framework for assessing the quality of OCR-derived historical election data. By prioritizing logical and accounting consistency, distinguishing between levels of severity, and clearly separating automatic corrections from manual review, the approach ensures both rigor and transparency.

The resulting audit log serves as a comprehensive diagnostic tool, enabling targeted data cleaning while preserving the integrity of the original dataset. Overall, the dataset is suitable for empirical analysis, provided that the identified high- and critical-severity issues are addressed through careful validation.